

# Retail Order Management Platform

## 1. EPIC

Build and automate CI workflow for Retail Order Management Platform

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## 2. FEATURES

### 2.1 Order Service API Development

#### 2.1.1 User Story

As a system, I want a health check API to ensure service reliability

##### Tasks

- Create Flask application
- Implement "/" home endpoint
- Implement "/health" endpoint
- Test health endpoint locally

#### 2.1.2 User Story

As a user, I want to check order status so that I know my order progress

##### Tasks

- Implement "/order-status" endpoint
  - Return order status response
  - Validate API response
  - Test endpoint locally
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### 2.2 Inventory Management API

#### 2.2.1 User Story

As a customer, I want to check inventory availability before placing an order

##### Tasks

- Implement "/inventory" endpoint
  - Return inventory availability response
  - Validate API response
  - Test endpoint locally
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## **2.3 Automated Testing**

### **2.3.1 User Story**

As a developer, I want automated tests to validate API functionality

#### **Tasks**

- Install pytest
  - Write test cases for "/" endpoint
  - Write test cases for "/health" endpoint
  - Write test cases for "/order-status" endpoint
  - Write test cases for "/inventory" endpoint
  - Run tests locally
  - Fix failing tests
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## **2.4 Repository Management (GitHub)**

### **2.4.1 User Story**

As a team member, I want version control to manage code changes efficiently

#### **Tasks**

- Create GitHub repository
  - Add project files (app.py, test\_app.py, requirements.txt)
  - Commit code with proper messages
  - Push code to main branch
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## **2.5 CI Pipeline Setup (GitHub Actions)**

### **2.5.1 User Story**

As a DevOps engineer, I want CI pipeline automation for continuous integration

#### **Tasks**

- Create .github/workflows directory
- Add python-app.yml workflow file
- Configure Python setup in pipeline
- Install dependencies in pipeline
- Run pytest in pipeline

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## **2.6 Build Verification and Validation**

### **2.6.1 User Story**

As a developer, I want to ensure the build passes successfully before deployment

#### **Tasks**

- Trigger pipeline on code push
- Monitor GitHub Actions workflow
- Verify all tests pass
- Fix pipeline errors if any
- Confirm successful build execution

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## **2.7 Application Monitoring**

### **2.7.1 User Story**

As a system admin, I want to monitor application health status

#### **Tasks**

- Use "/health" endpoint for monitoring
- Validate service availability
- Ensure correct response is returned